

Welcome to ENEL445

Engineering Optimisation

2025

Minimum-Distance Bridge Placement

B



A



Minimum-Distance Bridge Placement

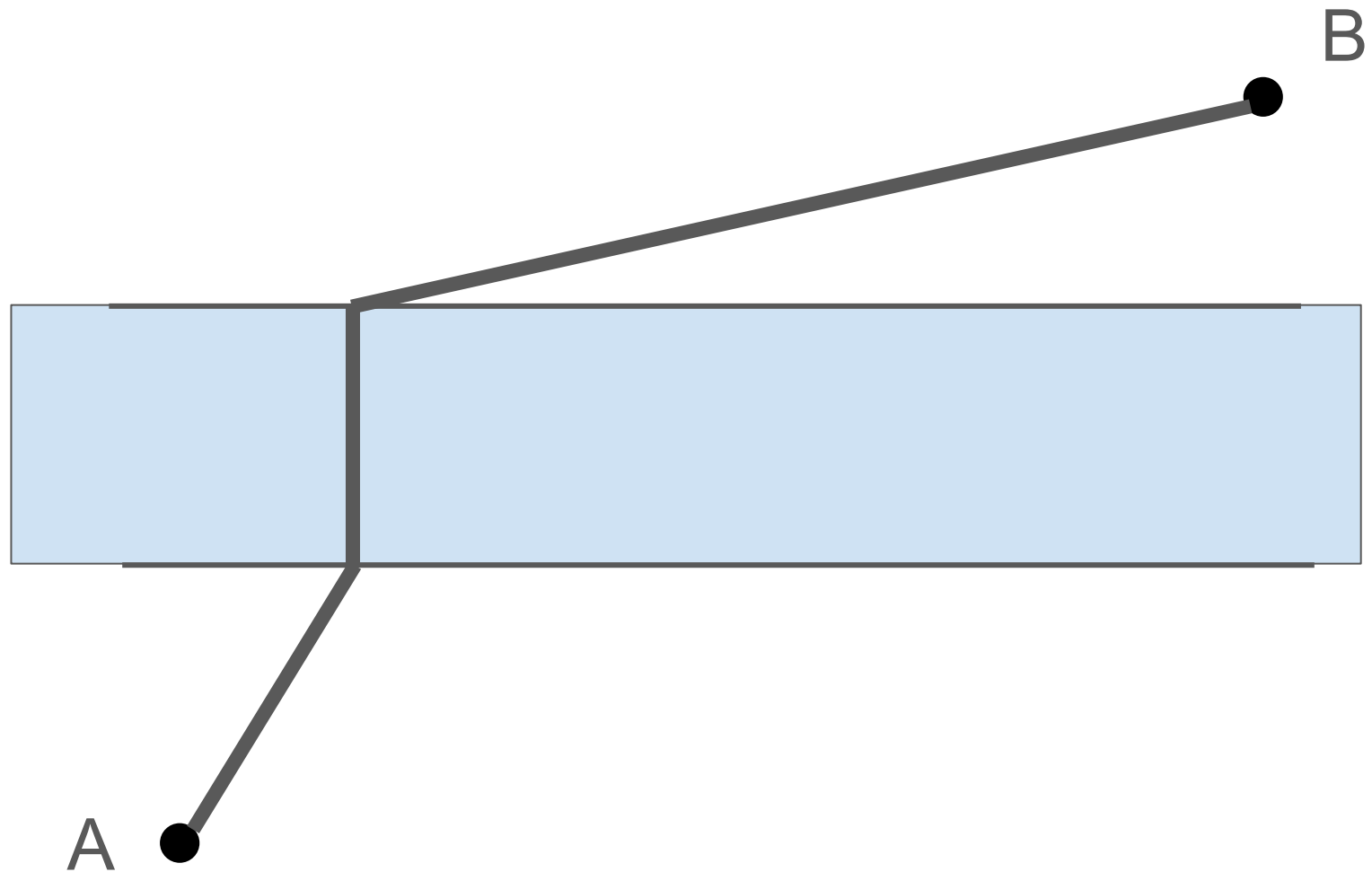
B



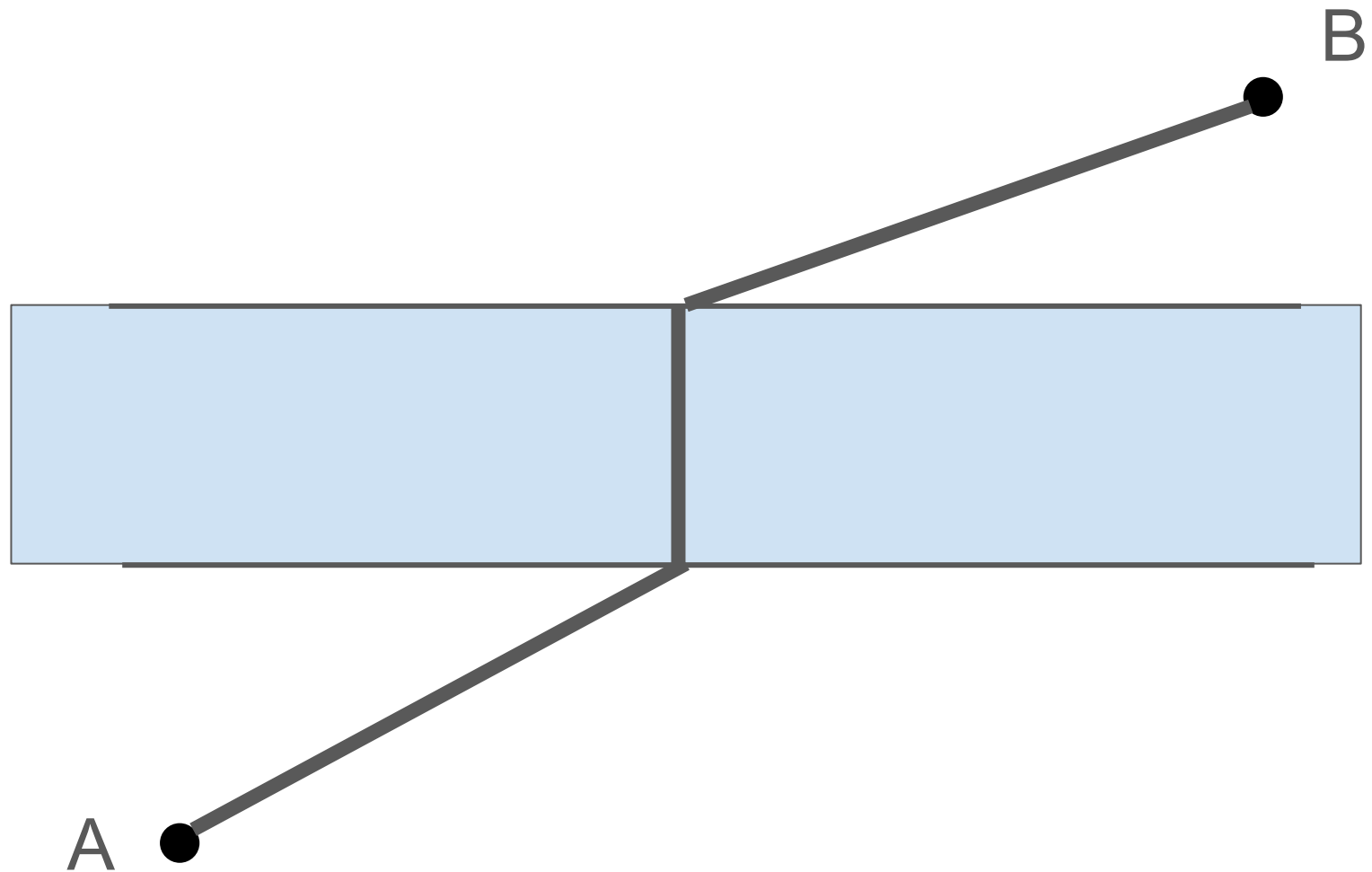
A



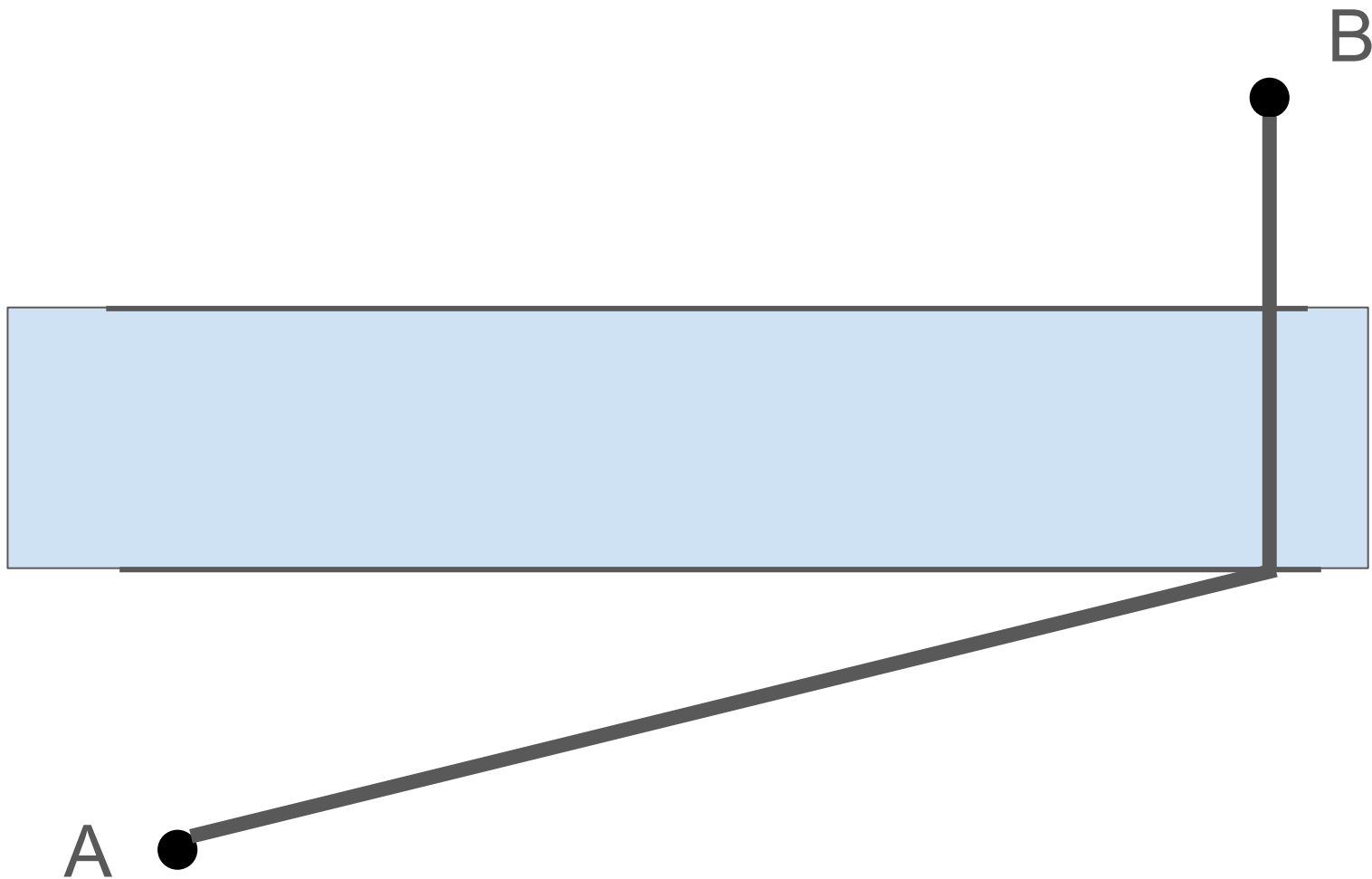
Minimum-Distance Bridge Placement



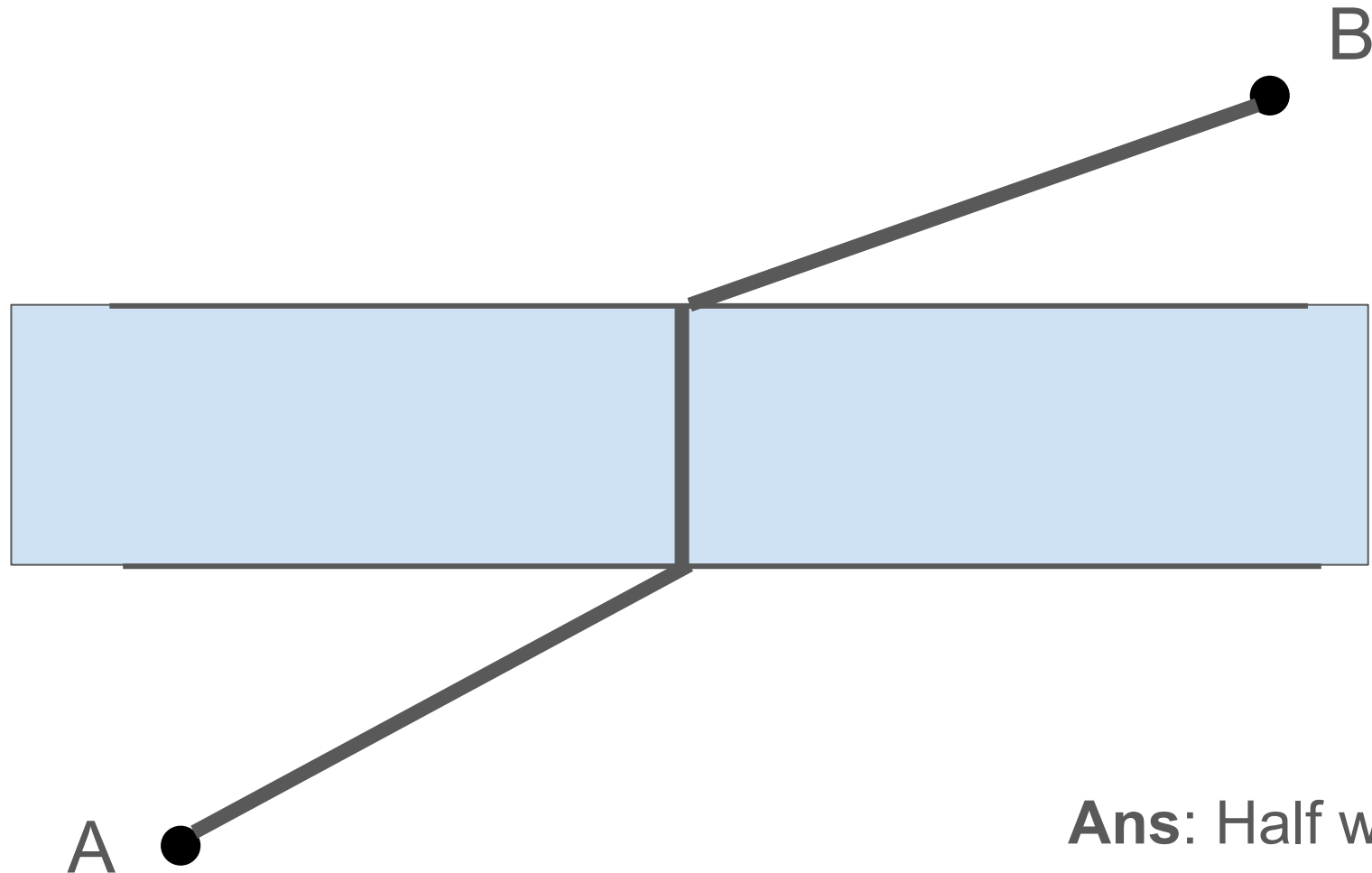
Minimum-Distance Bridge Placement



Minimum-Distance Bridge Placement

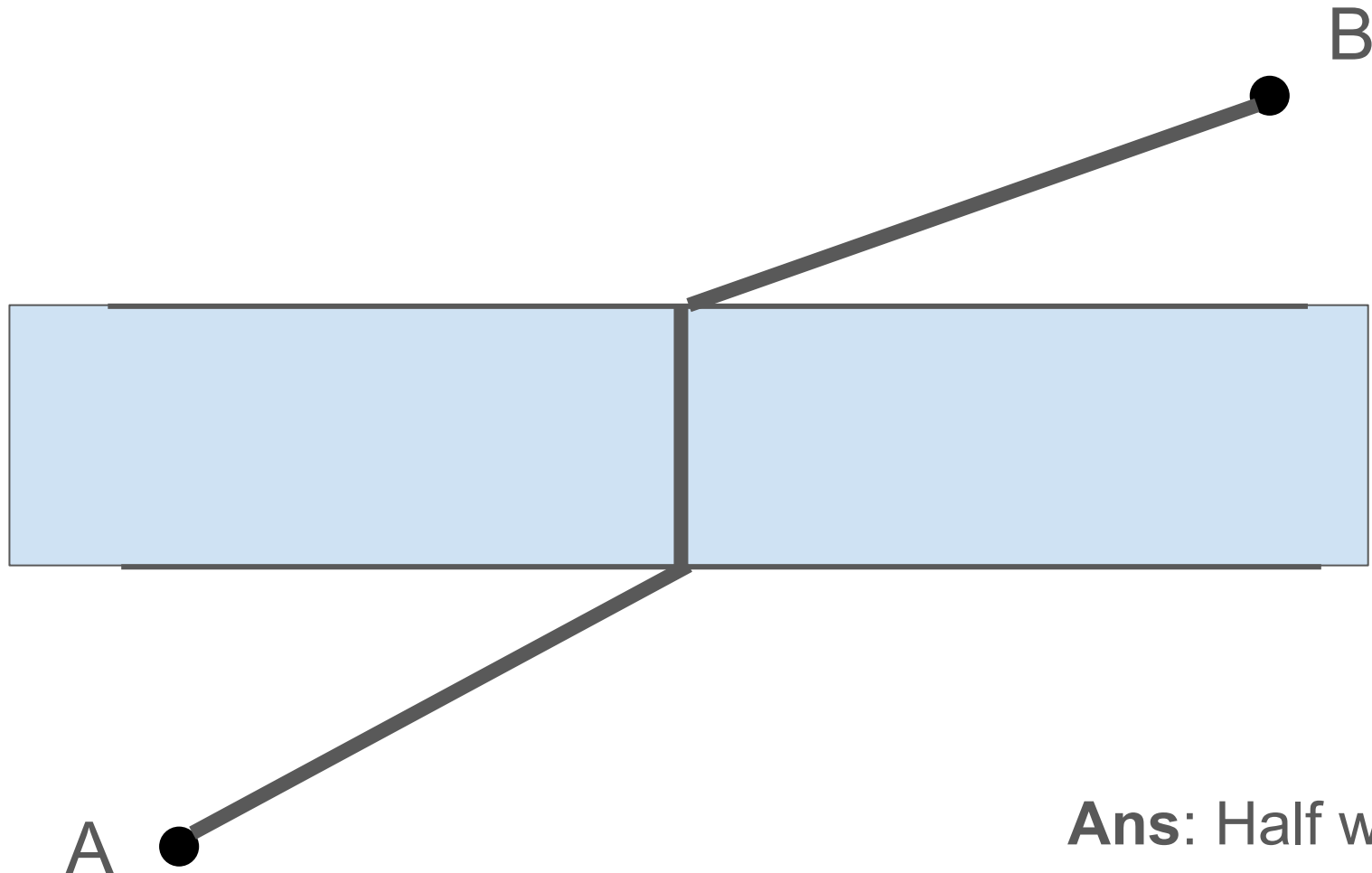


Minimum-Distance Bridge Placement



Ans: Half way

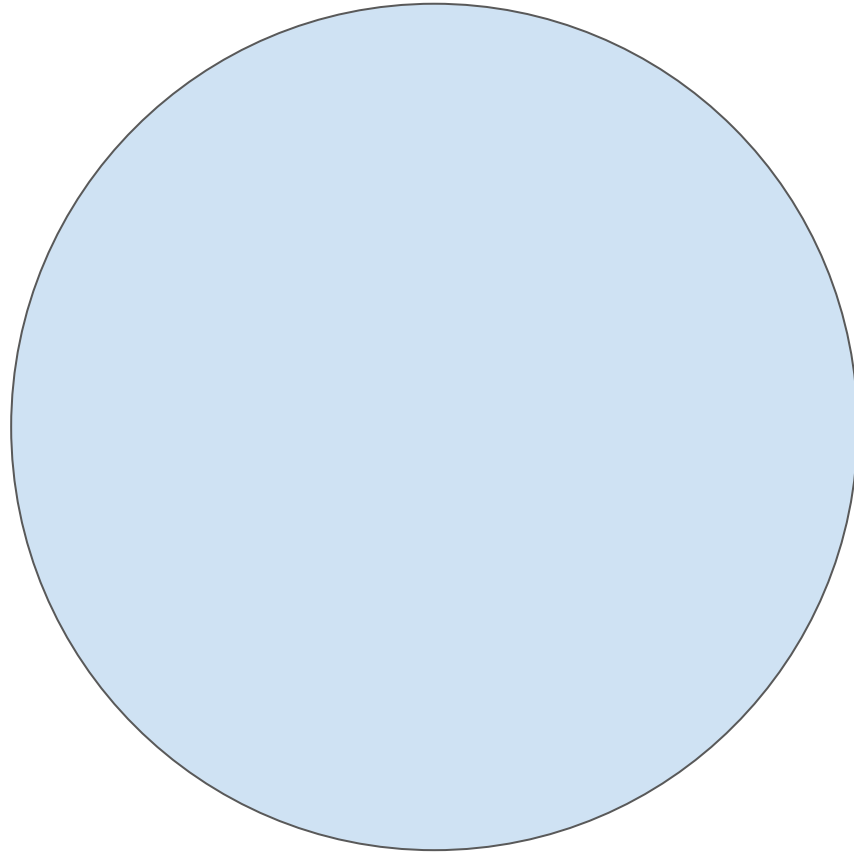
Minimum-Distance Bridge Placement



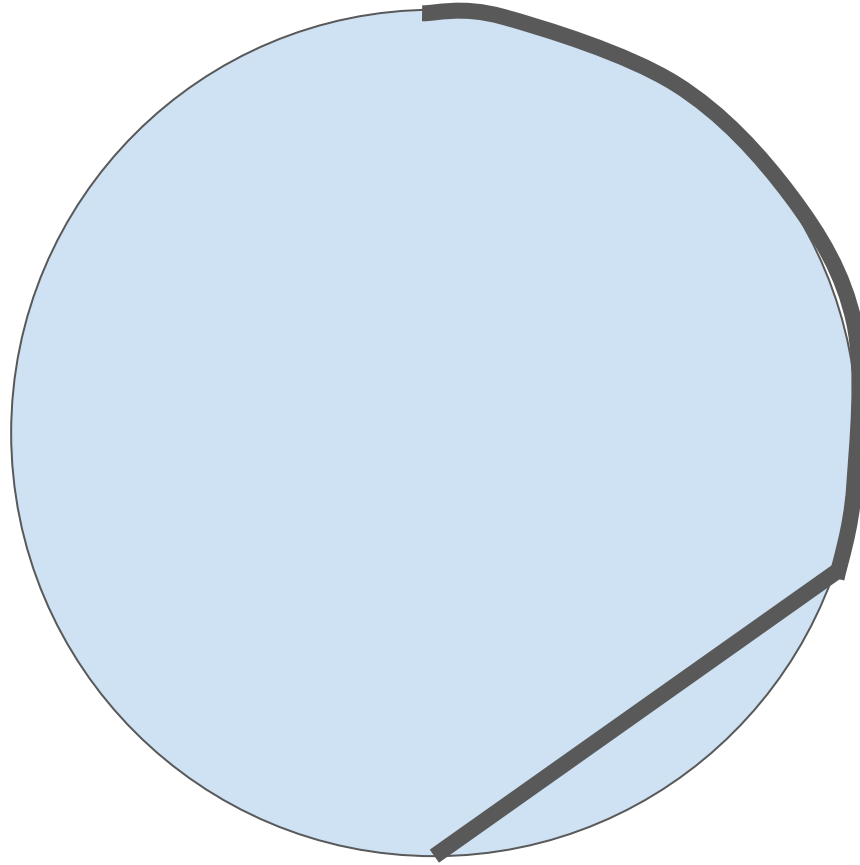
Ans: Half way

Homework Problem: Prove it!

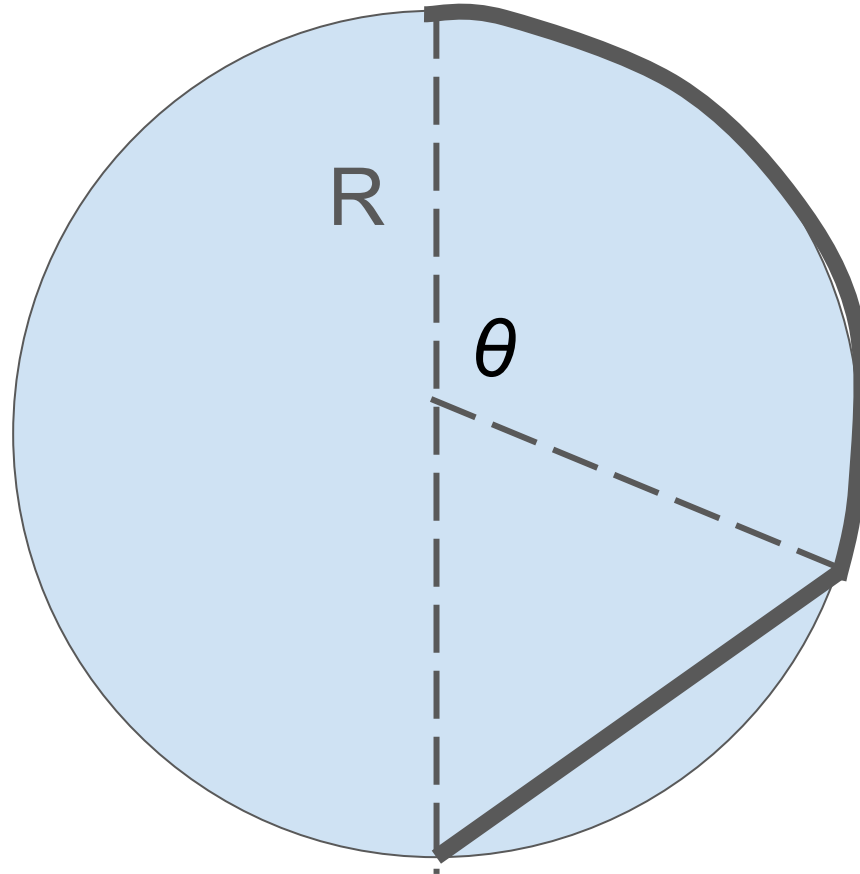
How to Cross a Circular Lake



How to Cross a Circular Lake

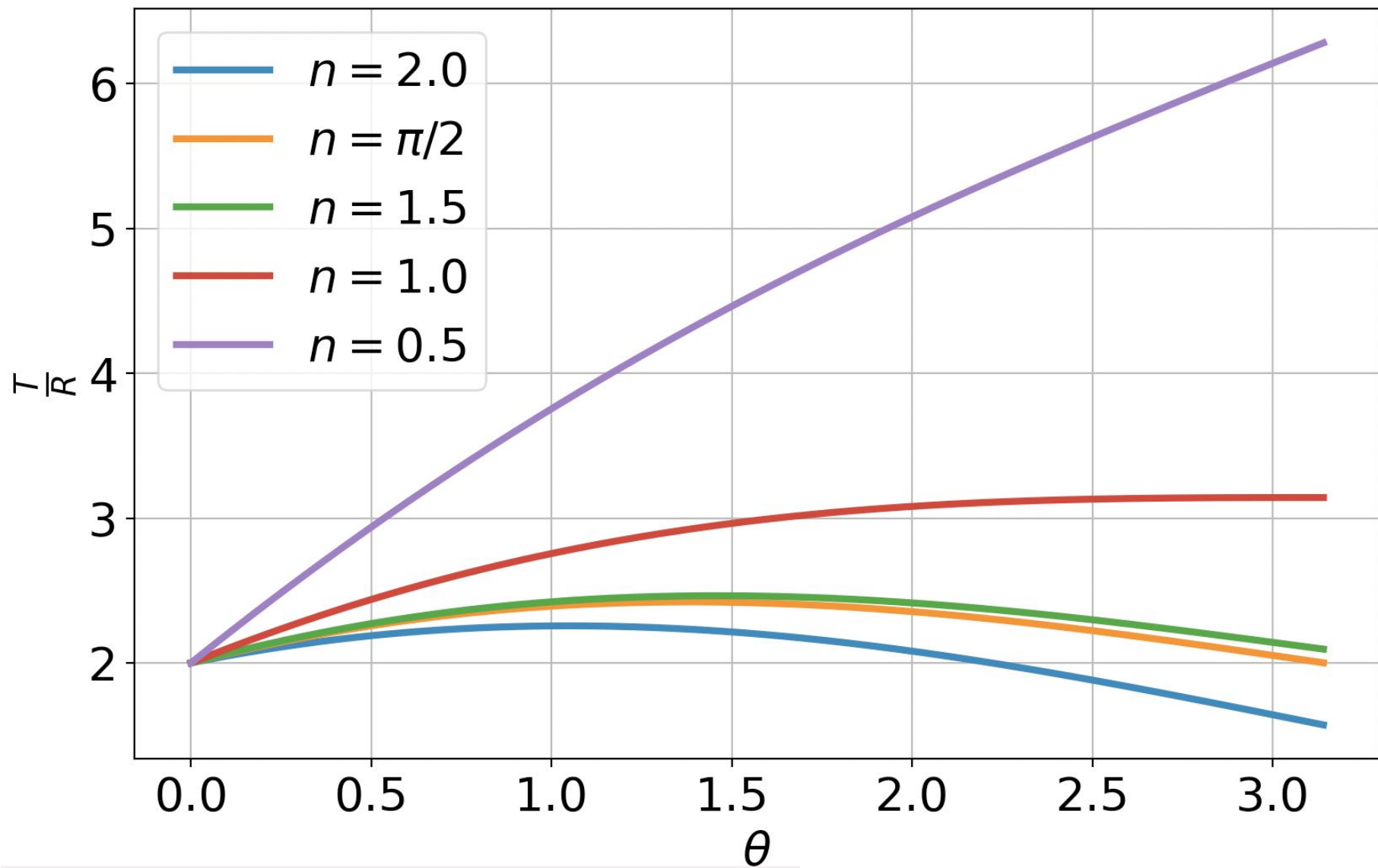


How to Cross a Circular Lake



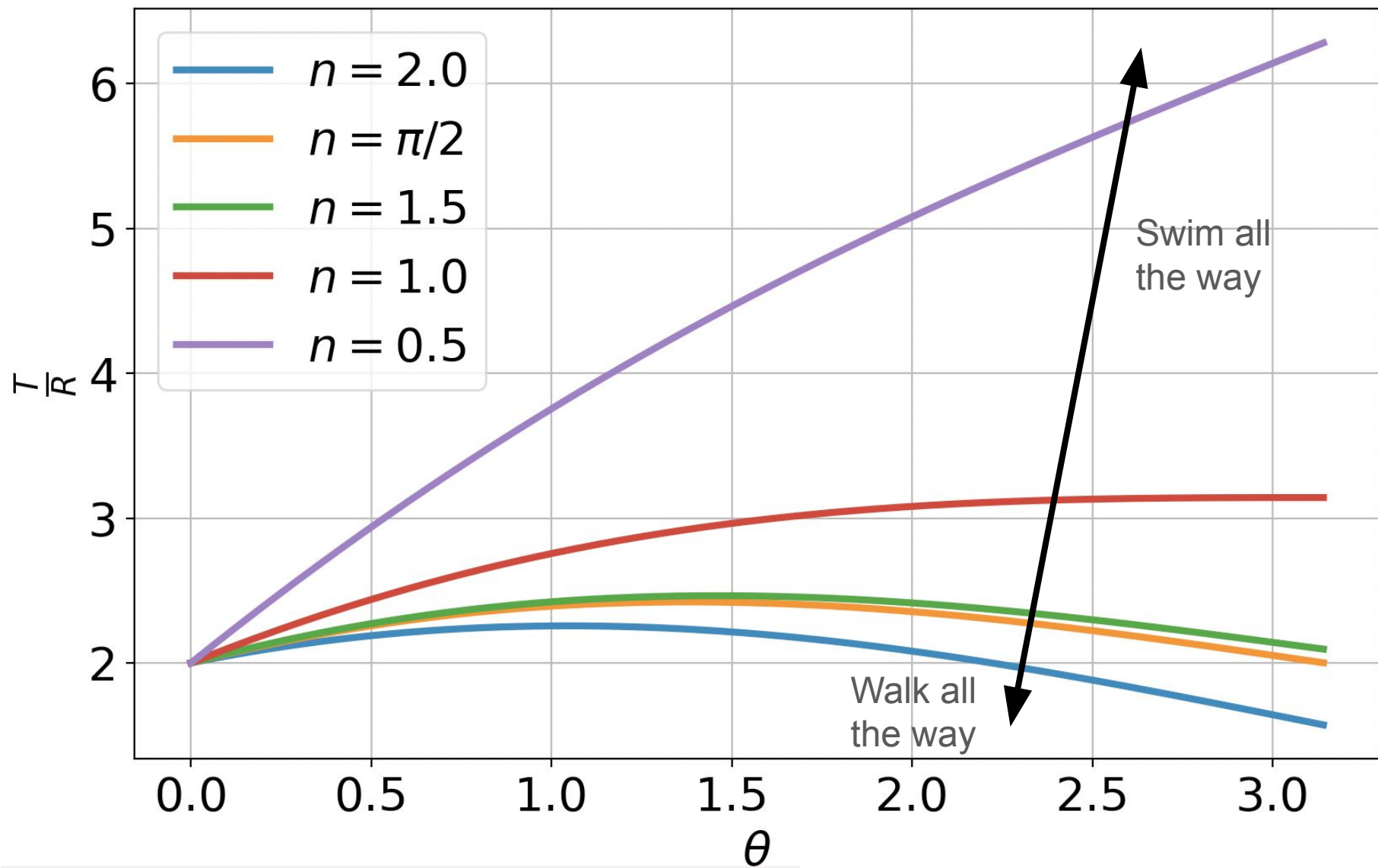
$$V_{\text{walk}} = nV_{\text{swim}}$$

$$T = 2R \cos \left(\frac{\theta}{2} \right) + \frac{R\theta}{n}$$



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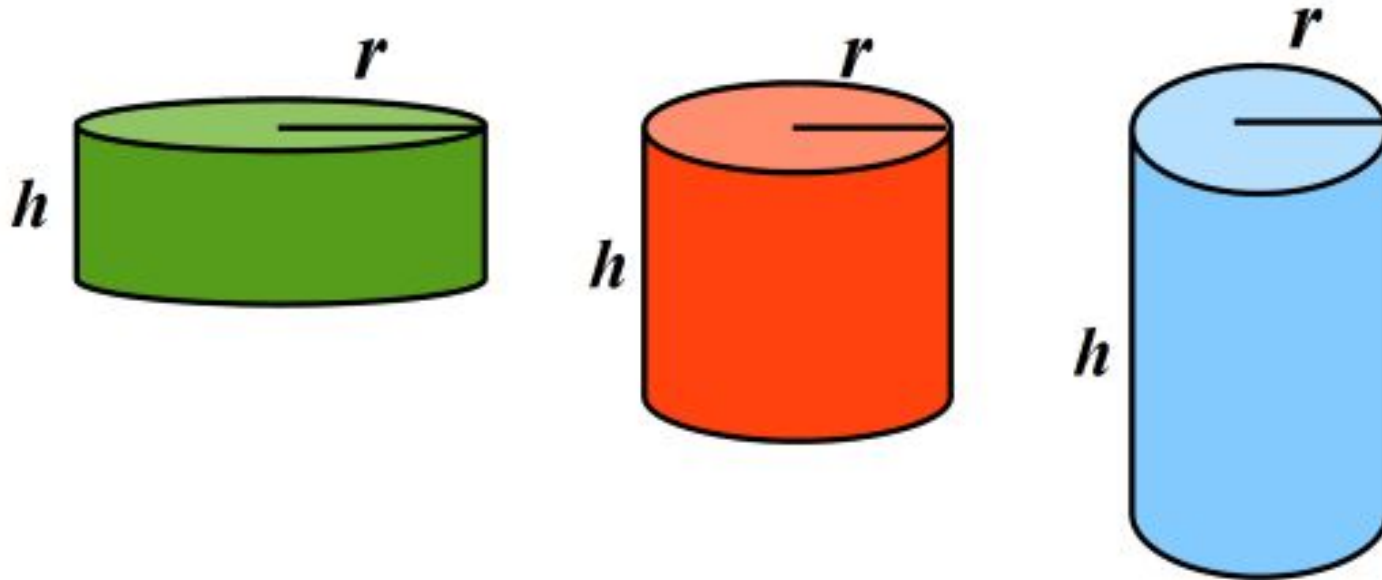


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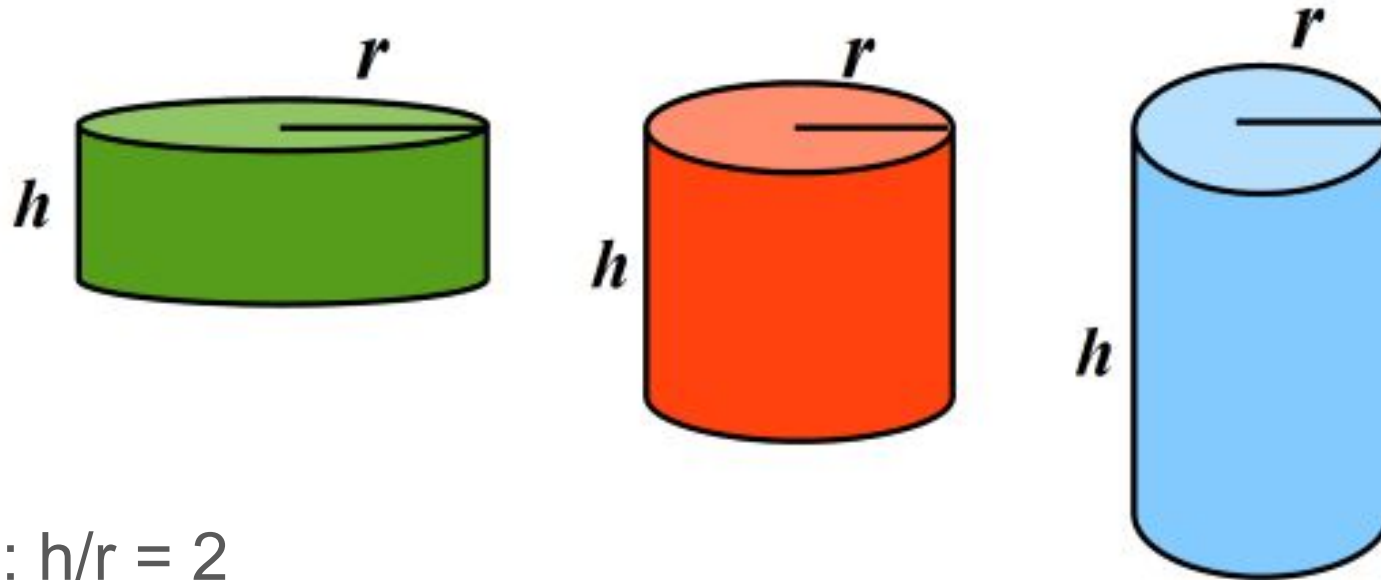
Minimum Surface Area

For cylinder with a fixed volume V , what ratio h/r gives the minimum surface area?



Minimum Surface Area

For cylinder with a fixed volume V , what ratio h/r gives the minimum surface area?



Ans: $h/r = 2$

Homework problem: For a cylinder, fix the surface area, maximise the volume

Minimum Surface Area





CLUB DEAL

MY DOG TENDER CHICKEN & TURKEY 680G

4.89

No Clubcard 6.25
7.19 per 1 KG

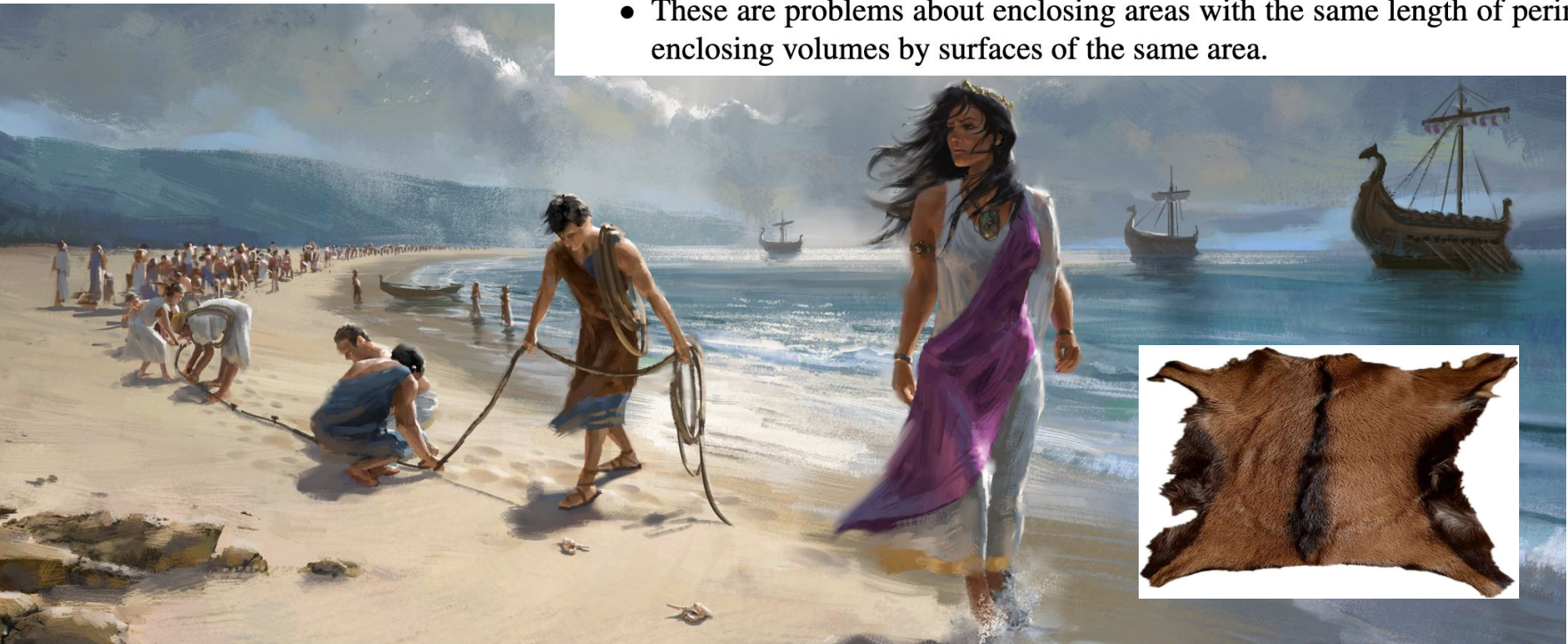
Only valid when you scan your Clubcard

Queen Dido

Dido's problem

- This is probably the oldest problem in the Calculus of Variations.
- Dido founded the city of Carthage, in Tunisia.
- According to legend, she arrived at the site with her entourage, a refugee from a power struggle with her brother in Tyre in the Lebanon.
- She asked the locals for as much land as could be bound by a bull's hide.
- She cut the hide into a long thin strip and bounded the maximum possible area with this.
- The maximum possible area bounded by a curve of fixed length is a circle. So the city of Carthage is circular in shape.
- Dido's problem is an example of what is called an *isoperimetric problem*.
- These are problems about enclosing areas with the same length of perimeter, or enclosing volumes by surfaces of the same area.

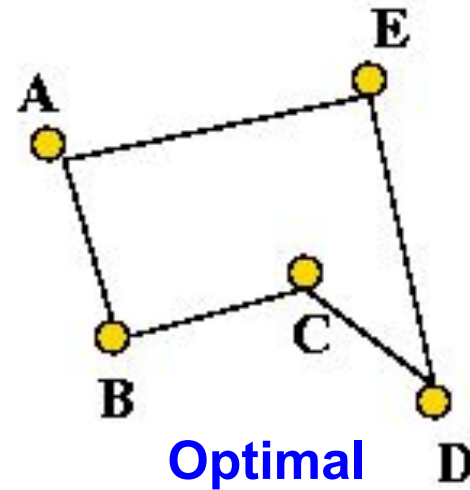
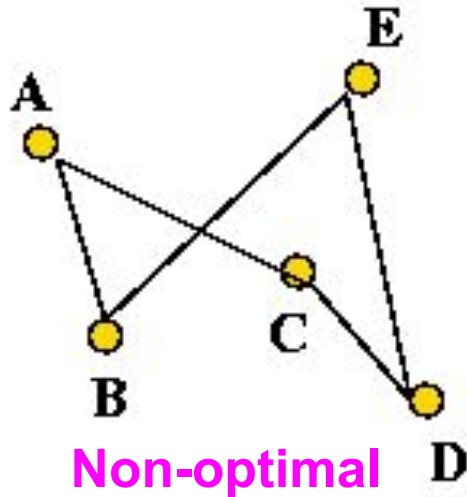
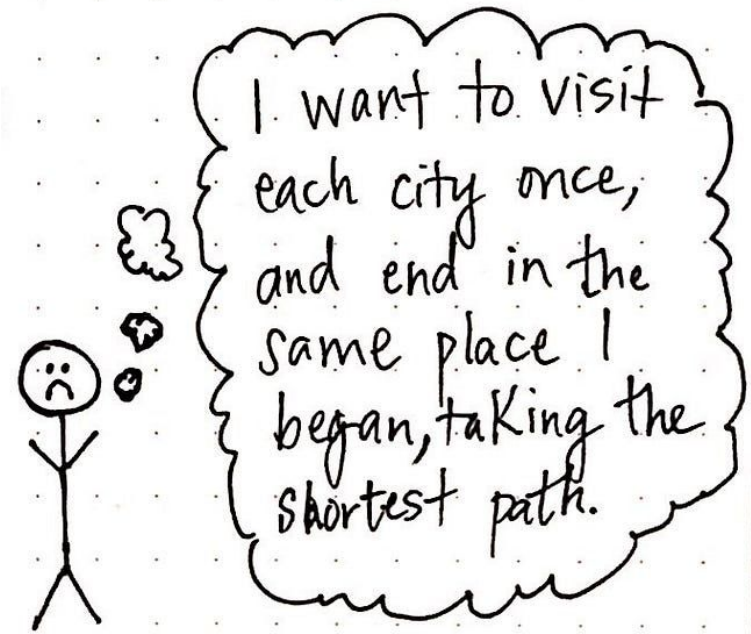
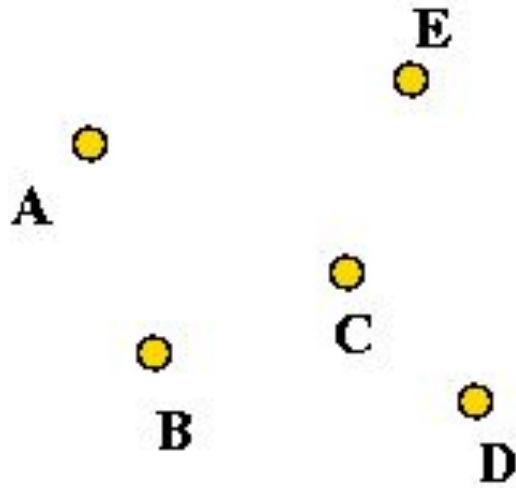
Isoperimetric Problem



Queen Dido



Travelling Salesman Problem





<https://towardsdatascience.com/how-i-saved-christmas-with-the-travelling-salesman-problem-3d85c190ed9d>

Organising the Best Birthday Party

Task Company	Catering	Decorations	Cake
A1 Festivities	\$80	\$40	\$70
Brilliant Birthdays	\$50	\$20	\$30
Cool Caterers	\$90	\$40	\$80

Organising the Best Birthday

Ans: Hungarian algorithm

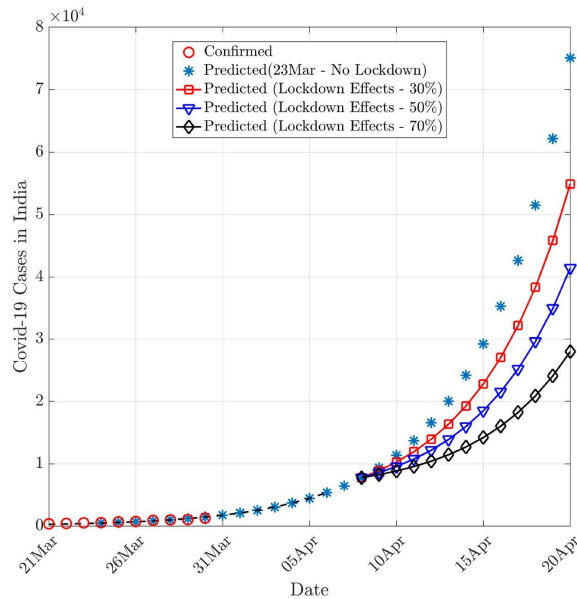
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Modern Optimisation

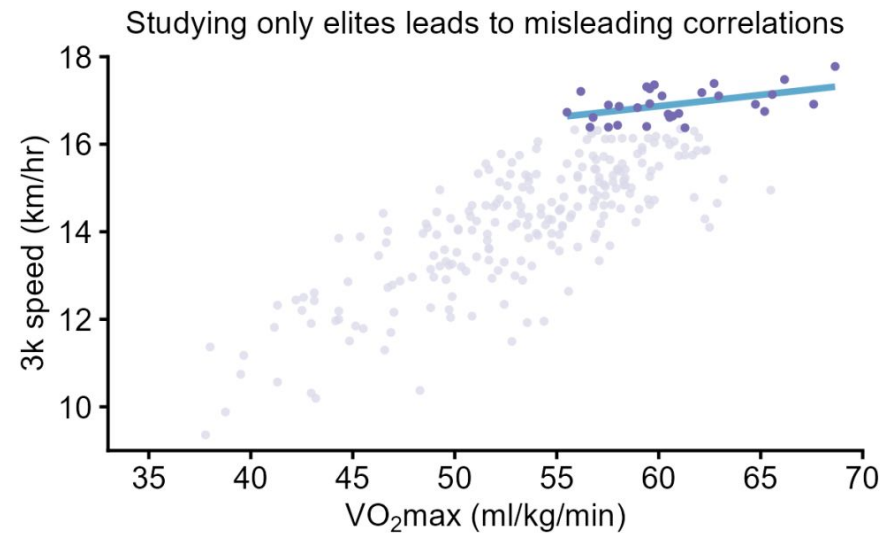
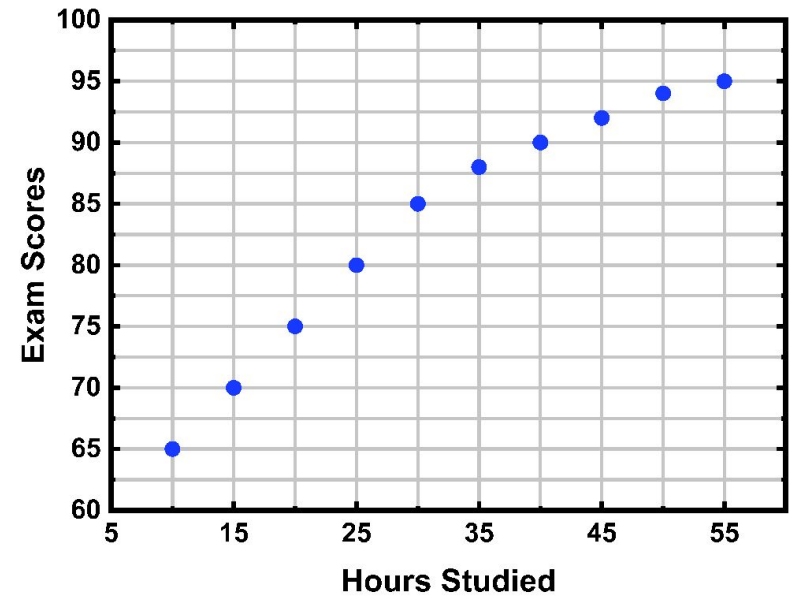
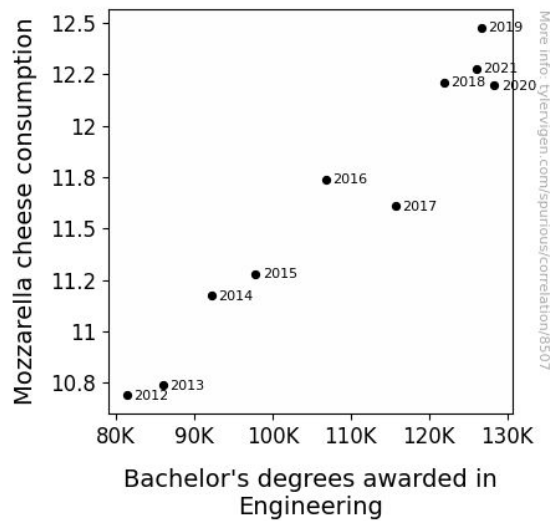
At the heart of today's science, engineering and technology: Finding the best **model** given **data**

- Fitting a curve
- Inverse problems - e.g. Imaging
- Model Predictive Control (MPC)
- Compressed sensing
- Machine learning

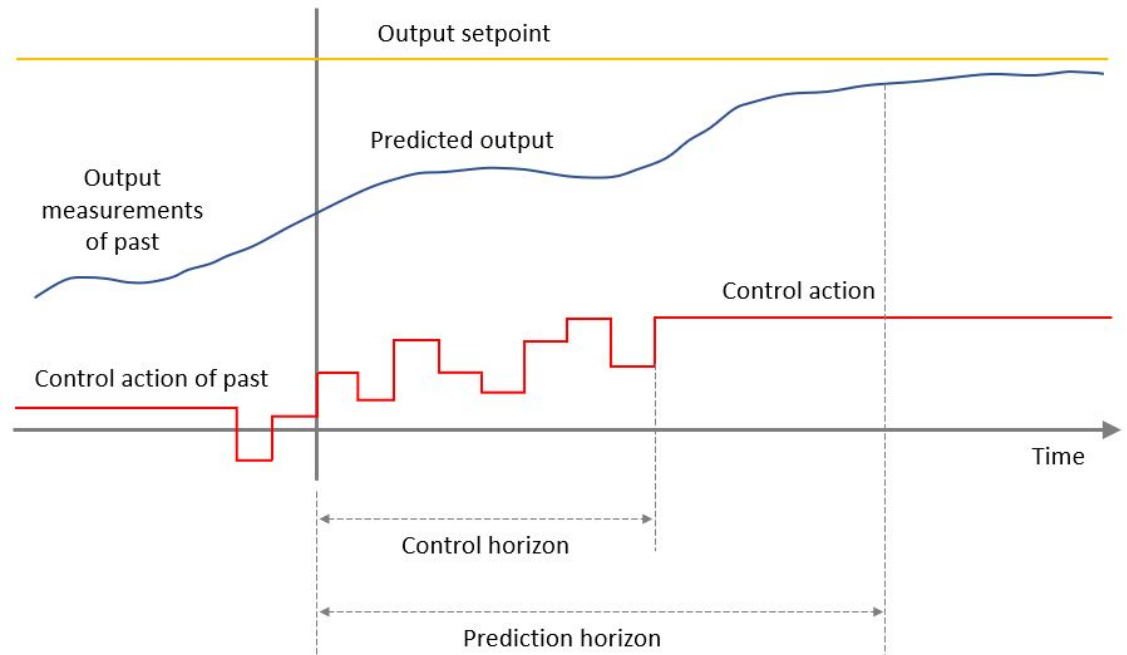
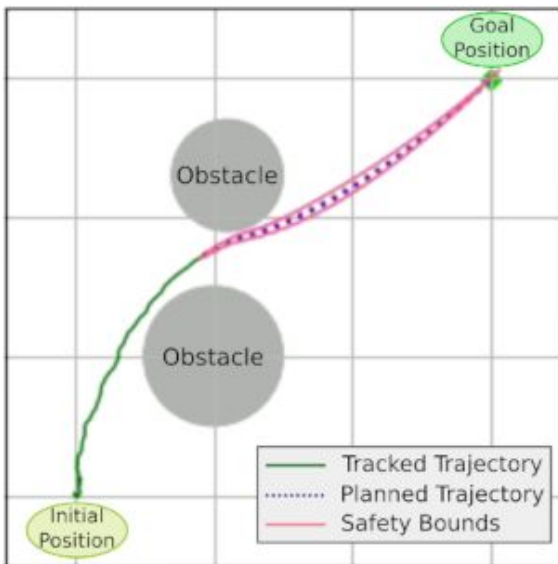
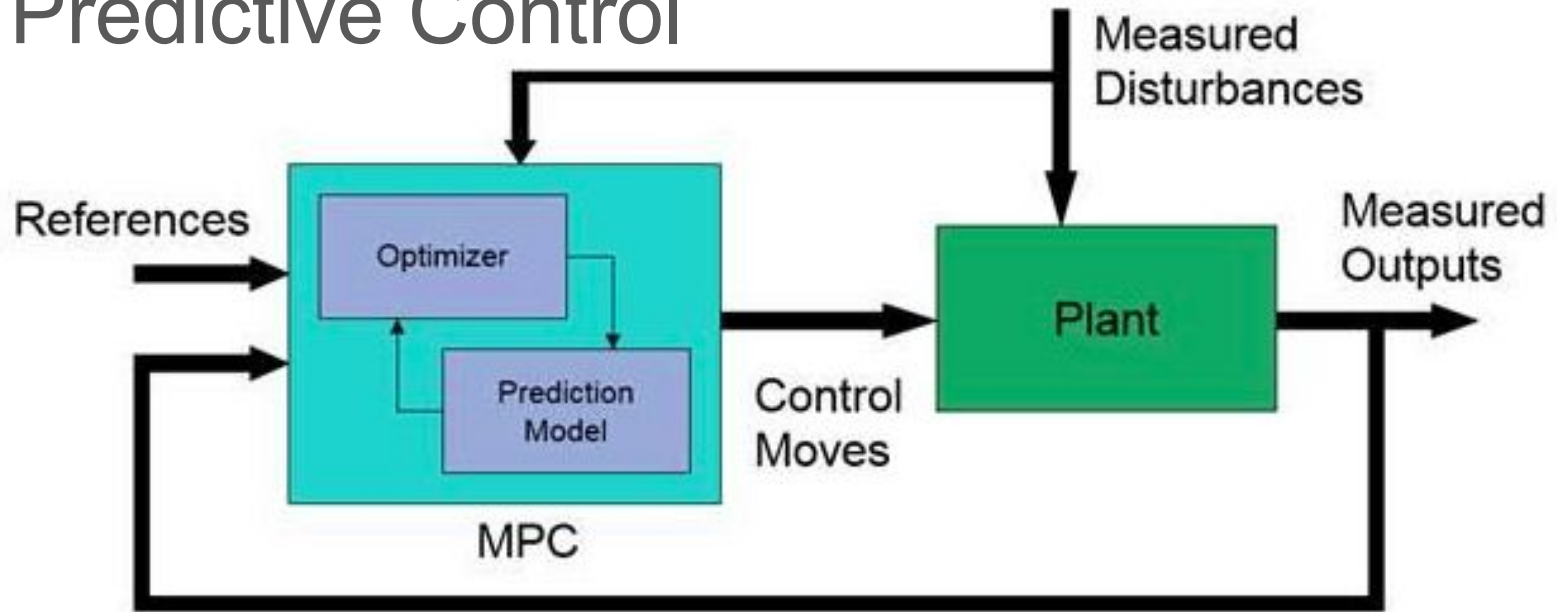
Fitting a Curve



Years 2012-2021, $r=0.974$, $p<0.01$



Model Predictive Control

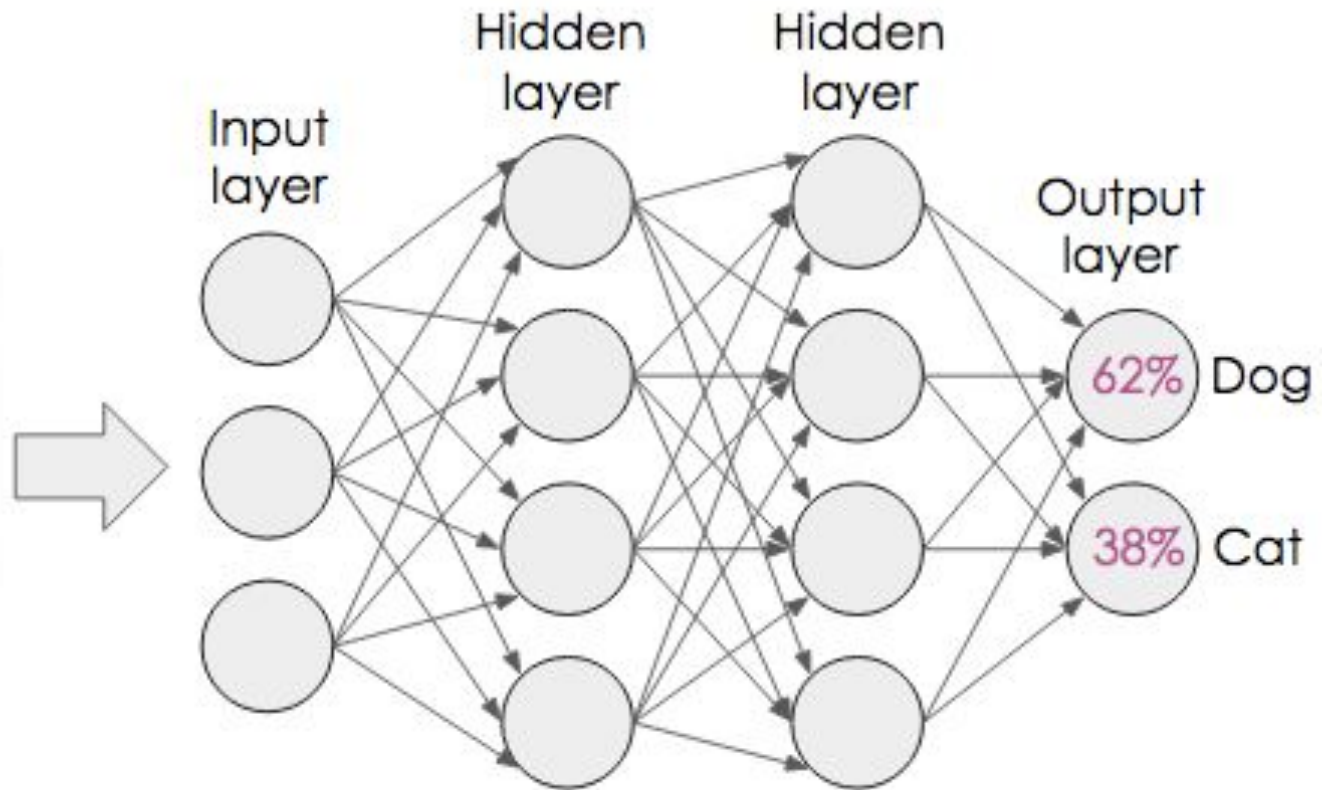


Compressed Sensing

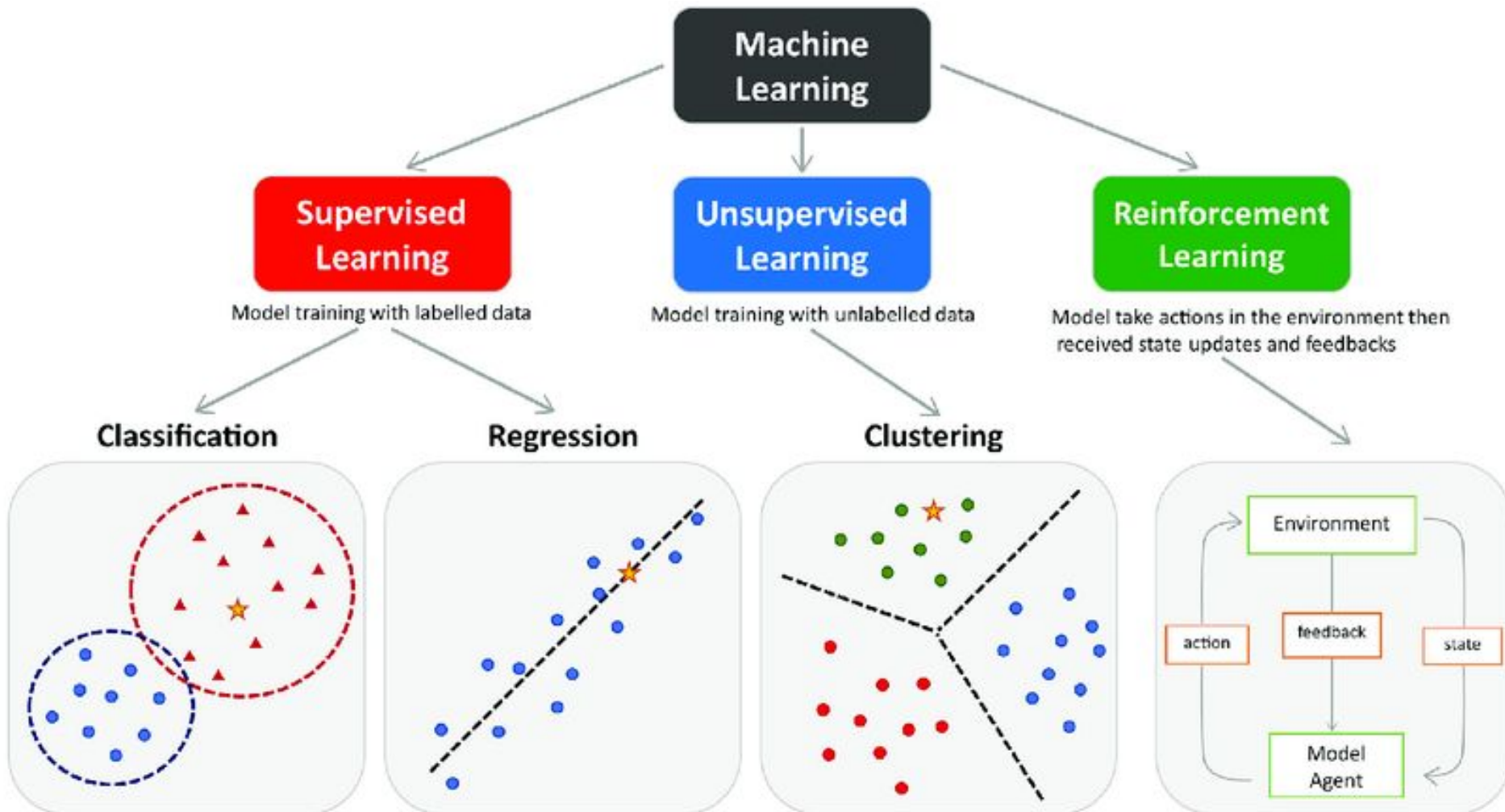
$$y = \Phi x$$

The diagram illustrates the Compressed Sensing equation $y = \Phi x$. The vector y (8x1) is shown on the left, the measurement matrix Φ (8x10) is in the center, and the signal vector x (10x1) is on the right. The vector y contains 8 colored elements: dark green, light purple, teal, magenta, blue, light green, red, and dark blue. The matrix Φ is an 8x10 grid of colored squares. The vector x contains 10 elements, with the 3rd element being green, the 4th being red, and the 9th being blue, while all other elements are white.

Machine Learning



Machine Learning



Past Projects

Previous years' projects:

- Optimised regenerative braking for UCM
- Computer AI for board games
- Rubik's cube, Sudoku solvers
- ENEL400 FYP allocations

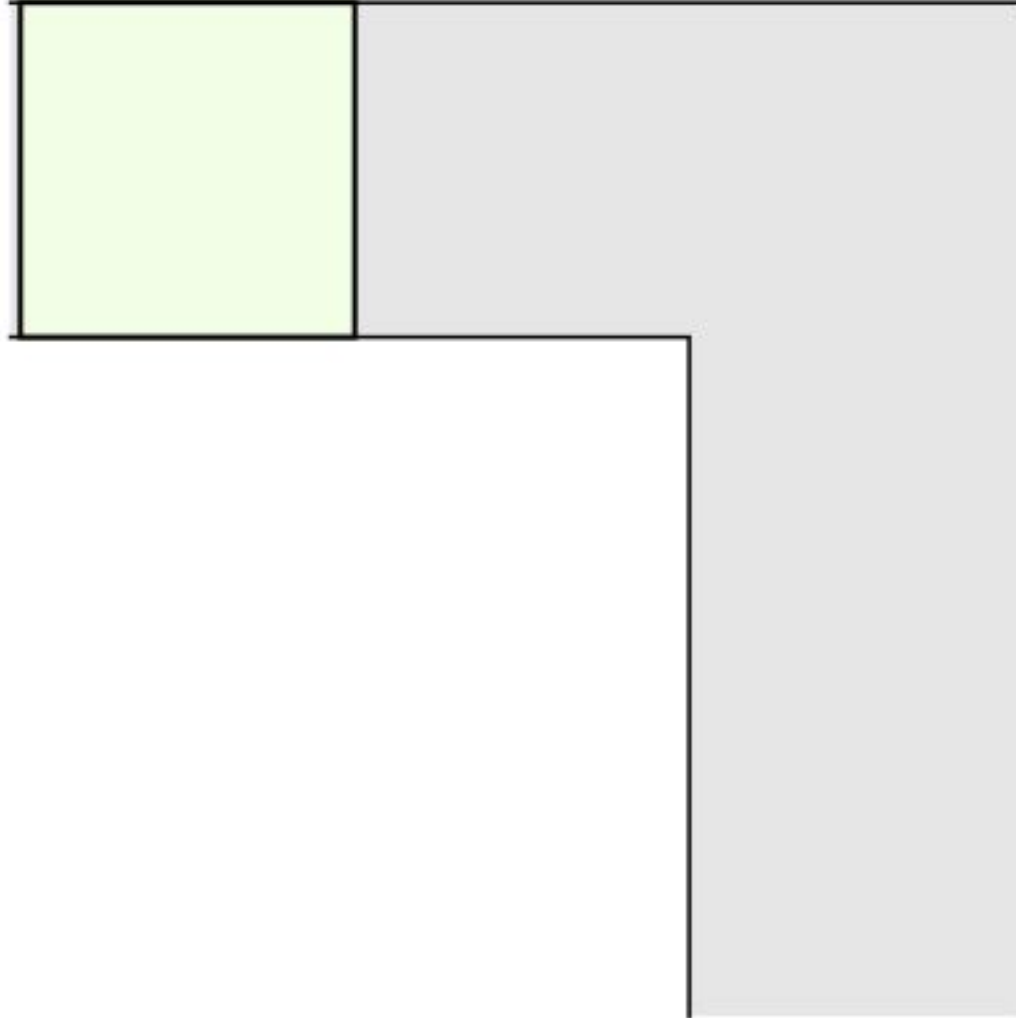
The Perfect Sofa



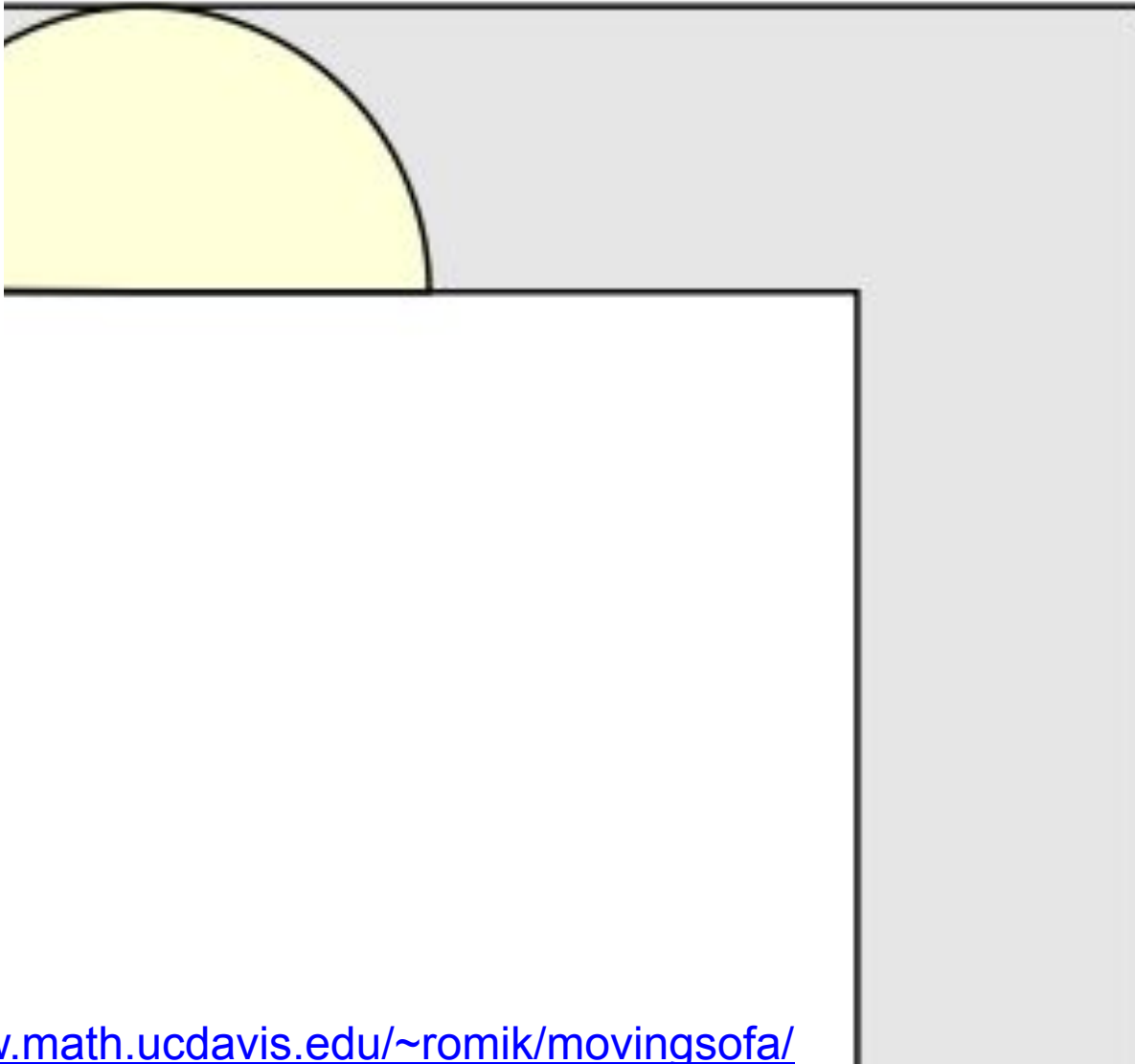
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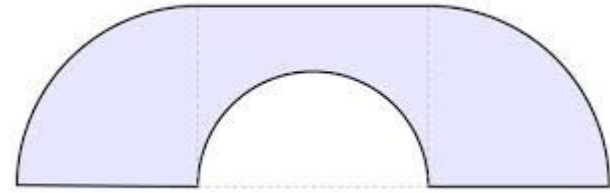


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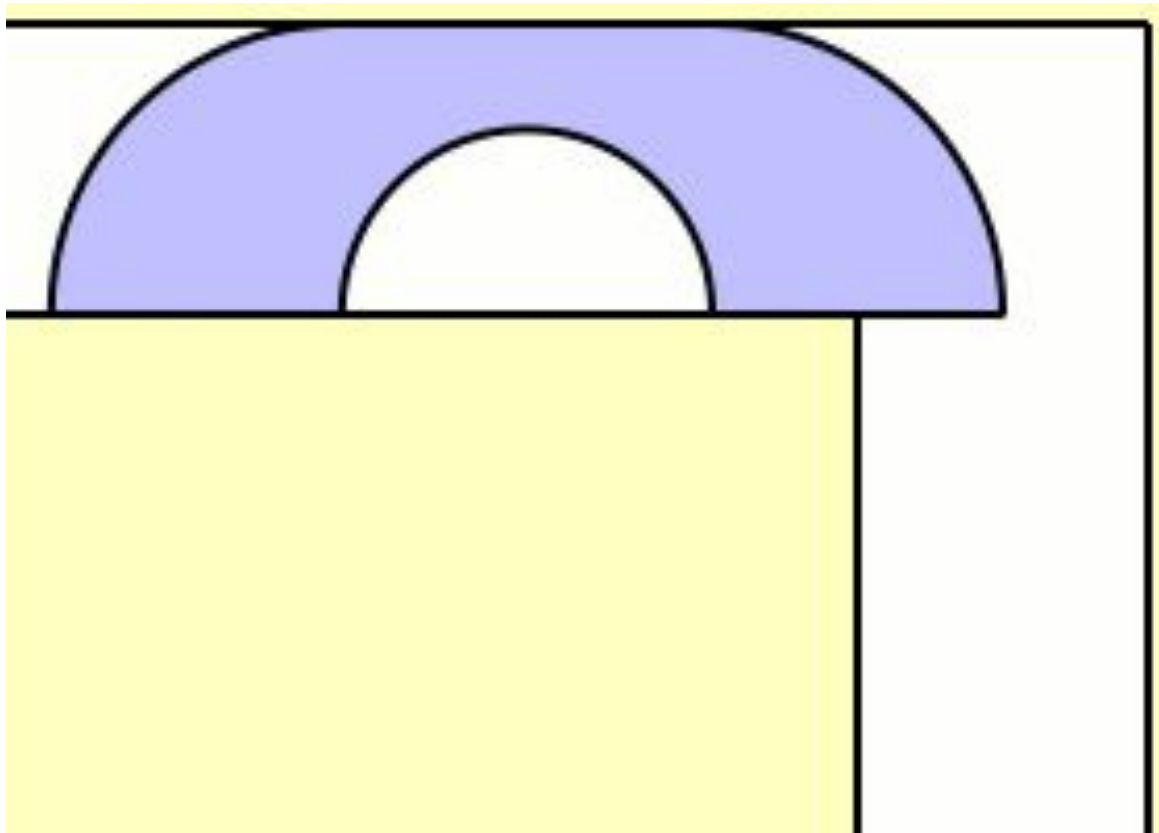


<https://www.math.ucdavis.edu/~romik/movingsofa/>

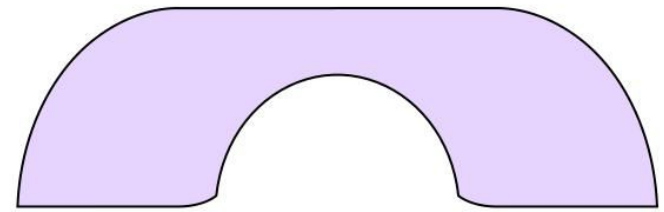
The Perfect Sofa?



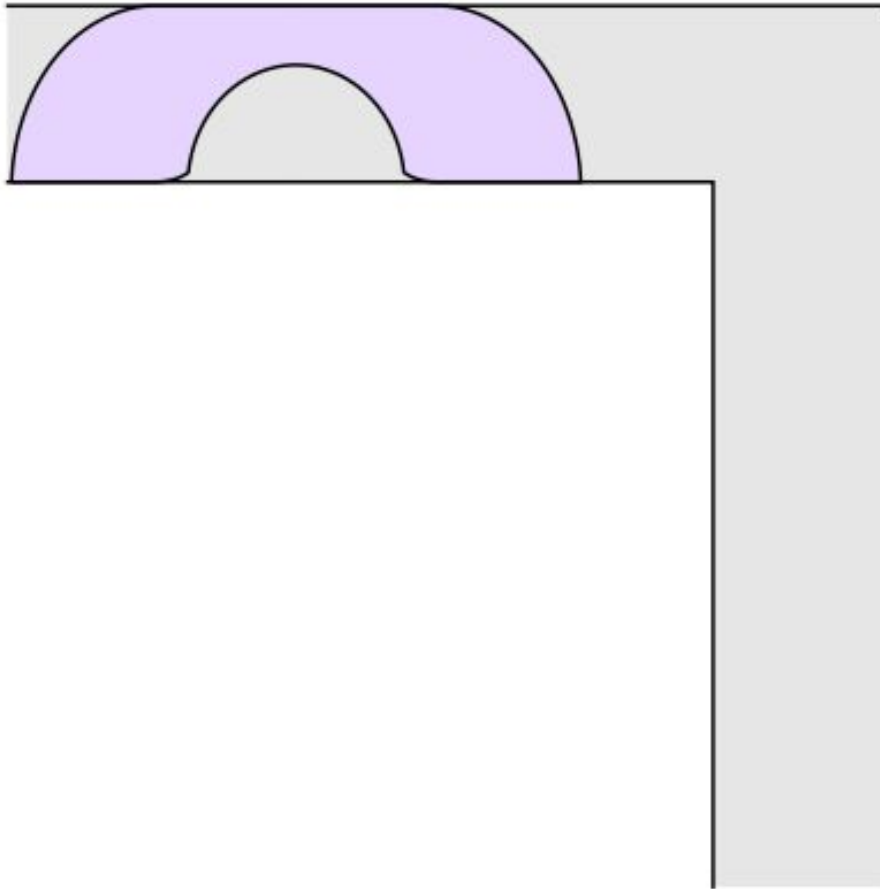
Ans: Hammersley sofa?



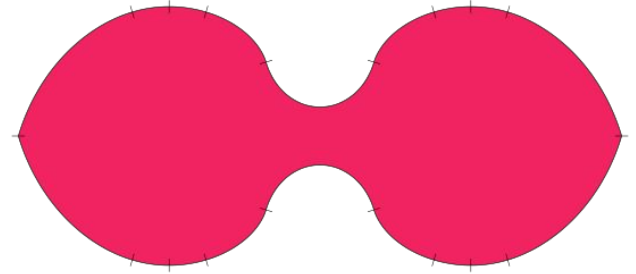
The Perfect Sofa?



Ans: Gerver sofa?



What if you have two corners?



Romik's ambidextrous sofa

